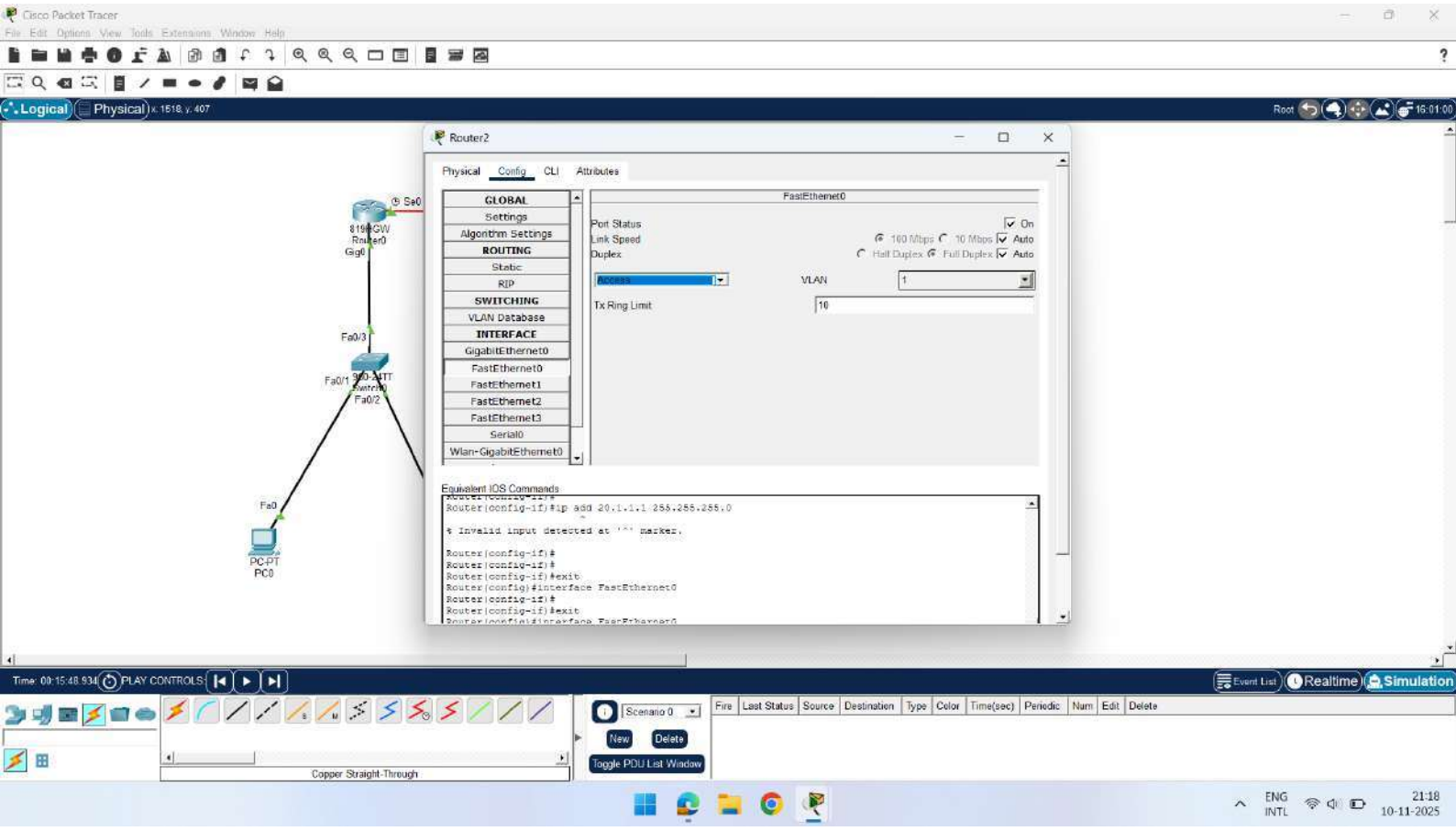

Cisco Packet Tracer
File Edit Options View Tools Extensions Window Help
Logical
Physical
Router2
Physical Config CLI Attributes
GLOBAL
Settings
Algorithm Settings
ROUTING
Static
RIP
SWITCHING
VLAN Database
INTERFACE
GigabitEthernet0
FastEthernet0
FastEthernet1
FastEthernet2
FastEthernet3
Serial0
Wlan-GigabitEthernet0
FastEthernet0
Port Status
On
Link Speed
100 Mbps 10 Mbps Auto
Duplex
Half Duplex Full Duplex Auto
Access
VLAN
1
Tx Ring Limit
10
Equivalent IOS Commands
Router(config-if)#ip add 20.1.1.1 255.255.255.0
% Invalid input detected at '^' marker.
Router(config-if)#
Router(config-if)#
Router(config-if)#exit
Router(config)#interface FastEthernet0
Router(config-if)#
Router(config-if)#exit
Se0
Gig0
Fa0/3
Fa0/1
Fa0/2
Fa0
PC-PT
PC0
Time: 00:15:48.934
PLAY CONTROLS:
Event List
Realtime
Simulation
Scenario 0
New
Delete
Toggle PDU List Window
Fire
Last Status
Source
Destination
Type
Color
Time(sec)
Periodic
Num
Edit
Delete
Copper Straight-Through
ENG
INTL
21:18
10-11-2025

Speed/Duplex Autonegotiation

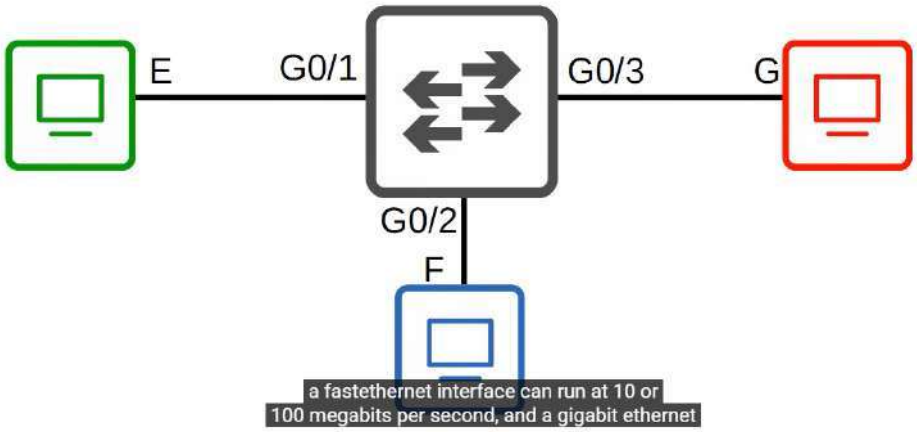
- Interfaces that can run at different speeds (10/100 or 10/100/1000) have default settings of **speed auto** and **duplex auto**.

- Interfaces 'advertise' their capabilities to the neighboring device, and they negotiate the best **speed** and **duplex** settings they are both capable of.

E = 10

F = 10/100

G = 10/100/1000



a fastethernet interface can run at 10 or
100 megabits per second, and a gigabit ethernet

Variable-Length Subnet Masks

- Until now, we have practiced subnetting used FLSM (Fixed-Length Subnet Masks).
- This means that all of the subnets use the same prefix length (ie. subnetting a class C network into 4 subnets using /26).

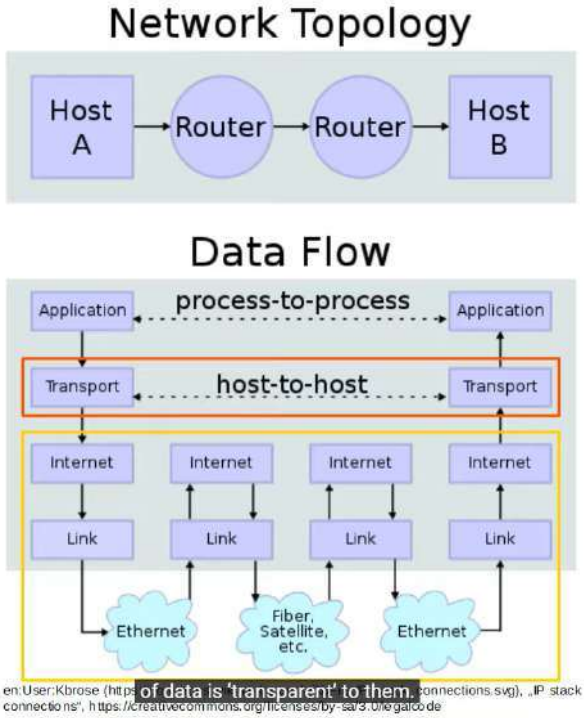
- VLSM (Variable-Length Subnet Masks) is the process of creating subnets of different sizes, to make your use of network addresses more efficient.

- VLSM is more complicated than FLSM, but it's easy if you follow the steps correctly.

So, that's what I want to teach you now,
the steps to subnet a network using VLSM.

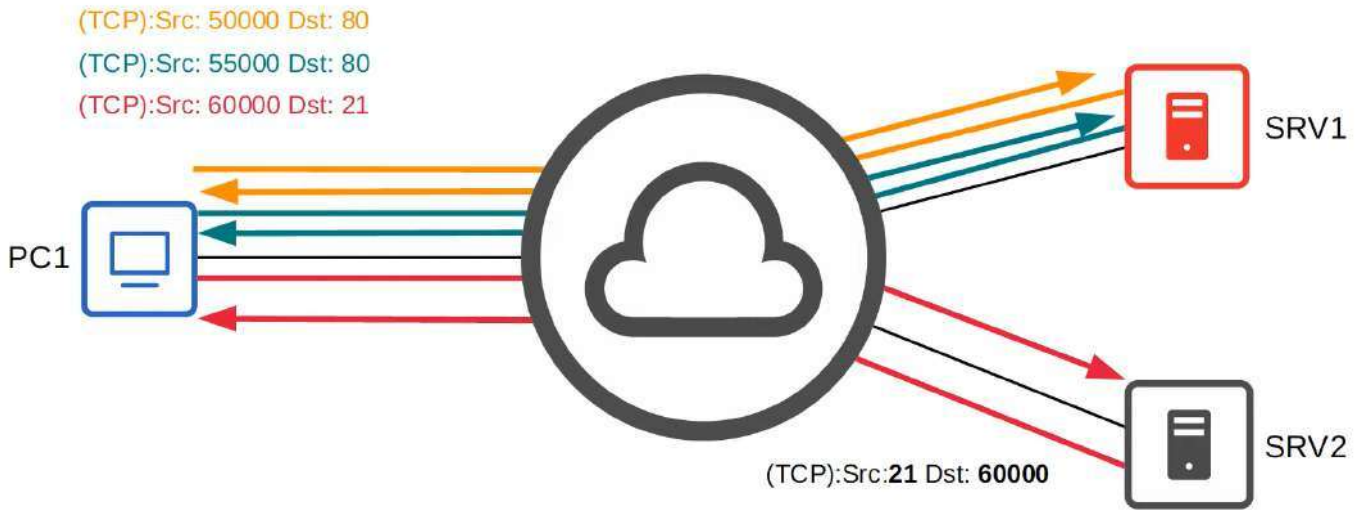
Functions of Layer 4 (Transport Layer)

- Provides transparent transfer of data between end hosts.



en:User:Kbrose (https of data is 'transparent' to them. connections.svg), „IP stack connections", https://creativecommons.org/licenses/by-sa/3.0/legalcode

Port Numbers / Session Multiplexing



These will tell PC1 that this communication is part of the same session.

7:07 / 33:58
Port Numbers: Session Multiplexing

Functions of Layer 4 (Transport Layer)

- Provides transparent transfer of data between end hosts.

- Provides (or doesn't provide) various services to applications:
  - → reliable data transfer
  - → error recovery
  - → data sequencing
  - → flow control

- Provides Layer 4 addressing (**port** numbers).

  ↪Identify the Application Layer protocol
  ↪Provides session multiplexing.

  ↪The following ranges have been designated by IANA (Internet Assigned Numbers Authority)
  Well-known port numbers: 0 – 1023
  Registered port numbers: 1024 – 49151
  Ephemeral/private/dynamic port numbers: 49152 – 65535

(TCP):Src: **50000** Dst: 80
(TCP):Src: **55000** Dst: 80
(TCP):Src: **60000** Dst: 21

earlier, reliable data transfer, error recovery,
data sequencing, and flow control, are provided

TCP (Transmission Control Protocol)

- TCP is connection-oriented.
  ↳ Before actually sending data to the destination host, the two hosts communicate to establish a connection. Once the connection is established, the data exchange begins.

- TCP provides reliable communication.
  ↳ The destination host must acknowledge that it received each TCP segment.
  ↳ If a segment isn't acknowledged, it is sent again.

- TCP provides sequencing.
  ↳ Sequence numbers in the TCP header allow destination hosts to put segments in the correct order even if they arrive out of order.

- TCP provides flow control.
  ↳ The destination host can tell the source host to increase/decrease the rate that data is sent.

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  ↳ The destination host can tell the source host to increase/decrease the rate that data is sent.

Here's the same summary I showed you before.

Free CCNA | TCP & UDP | Day 30 | CCNA 200-301 Complete Course
JEREMY'S IT LAB
Terminating Connections: Four-Way Handshake
FIN flag
ACK flag
FIN flag
ACK flag
PC1
SRV1
Finally, PC1 sends an ACK in response to SRV1's FIN, and the connection is terminated.
14:05 / 33:58
Terminating TCP Connections: 4-way Handshake

Comparing TCP & UDP

TCP	UDP
Connection-oriented	Connectionless
Reliable	Unreliable
Sequencing	No sequencing
Flow control	No flow control
Use for downloads, file sharing, etc	Used for VoIP, live video, etc

But don't forget, both TCP and UDP provide
Layer 4 addressing in the form of port numbers.

Port Numbers

TCP
- FTP data (20)
- FTP control (21)
- SSH (22)
- Telnet (23)
- SMTP (25)
- HTTP (80)
- POP3 (110)
- HTTPS (443)

UDP
- DHCP server (67)
- DHCP client (68)
- TFTP (69)
- SNMP agent (161)
- SNMP manager (162)
- Syslog (514)

TCP & UDP
- DNS (53)

Okay, to help you review I have added a 'portnumbers' tag to the Anki flashcards for these port

24:11 / 33:58
Important Port Numbers

Quiz 1

Which of the following is a well-known port number, as defined by IANA?

a) 1010

b) 2001

c) 4023

d) 65000

↳The following ranges have been designated by IANA (Internet Assigned Numbers Authority)
Well-known port numbers: 0 – 1023
Registered port numbers: 1024 – 49151
Ephemeral/private/dynamic port numbers: 49152 – 65535

Answers B and C are in the Registered range, and D is in the Ephemeral range.

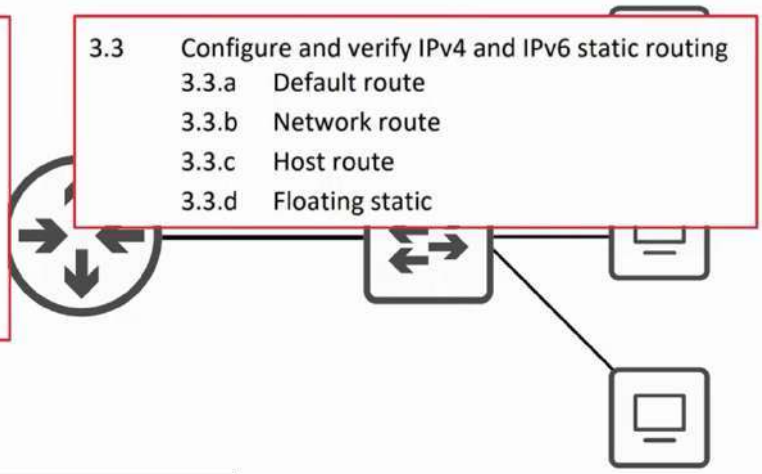
CCNA 200-301 Day 31

IPv6 Part 1

1.8 Configure and verify IPv6 addressing and prefix

1.9 Compare IPv6 address types
- 1.9.a Global unicast
- 1.9.b Unique local
- 1.9.c Link local
- 1.9.d Anycast
- 1.9.e Multicast
- 1.9.f Modified EUI 64

3.3 Configure and verify IPv4 and IPv6 static routing
- 3.3.a Default route
- 3.3.b Network route
- 3.3.c Host route
- 3.3.d Floating static



I considered trying to fit all of this into a single video, but have decided to split

Hexadecimal

- Binary / Base 2 / 0b
  0, 1

10 ← Is that decimal 10?
Or binary 10 (=decimal 2)?
Or hexadecimal 10 (=decimal 16)?

0b10

- Decimal / Base 10 / 0d
  0, 1, 2, 3, 4, 5, 6, 7, 8, 9

- Hexadecimal / Base 16 / 0x
  0, 1, 2, 3, 4, 5, 6, 7, 8, 9, A, B, C, D, E, F

B, C, D, E, and F are used as well.

4:43 / 39:25
Hexadecimal review

Hexadecimal

Decimal	Binary	Hexadecimal
0	0000	0
1	0001	1
2	0010	2
3	0011	3
4	0100	4
5	0101	5
6	0110	6
7	0111	7
8	1000	8
9	1001	9

Decimal	Binary	Hexadecimal
10	1010	A
11	1011	B
12	1100	C
13	1101	D
14	1110	E
15	1111	F

Here's a chart comparing the three, from 0 up to decimal 15.

Binary → Hexadecimal 1

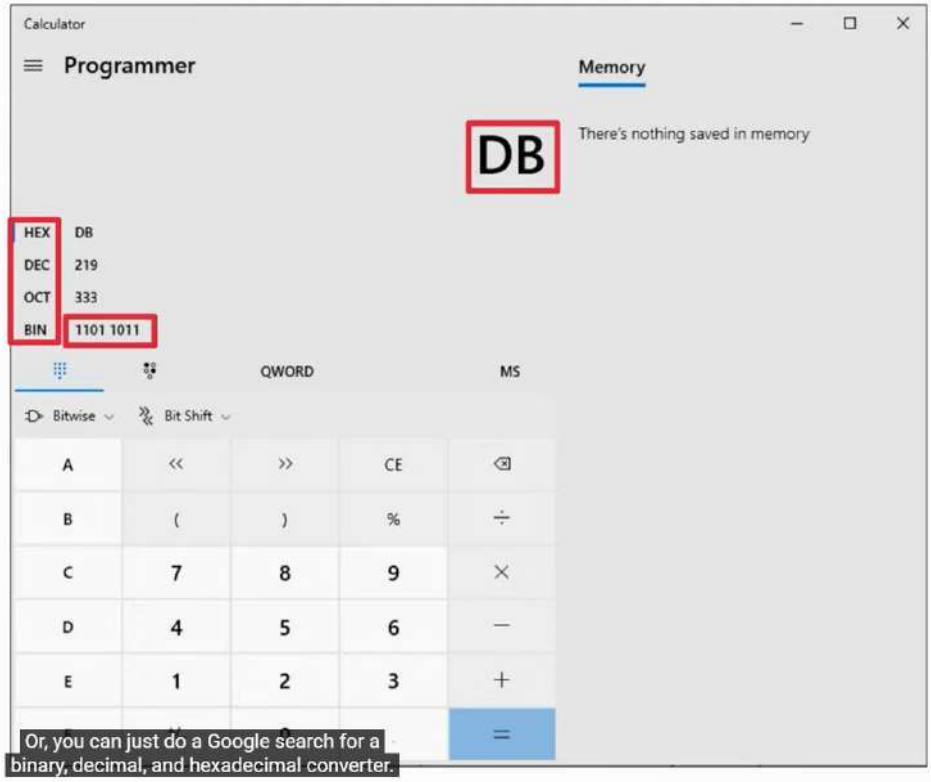
0b11011011 = 0x??

0b1101 0b1011

0d13 0d11

0xD 0xB

0b11011011 = 0xDB



Hexadecimal → Binary 1

0xEC = 0b??

0xE 0xC

Split up the hexadecimal digits

0d14 0d12

Convert each hexadecimal digit to decimal

0b1110 0b1100

Convert each decimal number to binary

0xEC = 0b11101100

That's the answer

And that's the answer, hexadecimal EC is equal to binary 1110 1100.

Why IPv6?

- IPv4 address assignments are controlled by IANA (Internet Assigned Numbers Authority)
- IANA distributes IPv4 address space to various RIRs (Regional Internet Registries), which then assign them to companies that need them.

On 24 September 2015 ARIN declared exhaustion of the ARIN IPv4 addresses pool.

On 21 August 2020, LACNIC announced that it had made its final IPv4 allocation.

Let's finally get into the specifics.

By Rr.gif: DorkBlankMap-World6,_compact.svg: Caruckguy et al.derivative work: Sémhur (talk) - Rr.gifBlankMap-World6,_compact.svg, CC BY-SA 3.0, https://commons.wikimedia.org/w/index.php?curid=5810575

IPv6

- An IPv6 address is **128 bits**.
- 4*the bits of an IPv4 address = 4*the number of possible addresses? NO
- Every additional bit **doubles** the number of possible addresses.
- There are 340,282,366,920,938,463,463,374,607,431,768,211,456 IPv6 addresses.
  There are ……..4,294,967,296 IPv4 addresses.
- Example IPv6 address in binary:
  00100000000000010000110110111000010110010001011111101010101111010110010101100001000010111111010101100100100101101010110011011110 1

  ↳~~32.1.13.184.89.23.234.189.101.98.23.234.201.45.89.189~~

  ↳2001:0DB8:5917:EABD:6562:17EA:C92D:59BD /64

1 2 3 4 5

In addition, there are a couple methods to shorten IPv6 addresses to make them simpler.

Shortening (abbreviating) IPv6 addresses

- **Leading 0s** can be removed

2001:0DB8:000A:001B:20A1:0020:0080:34BD

↓

2001:DB8:A:1B:20A1:20:80:34BD

- **Consecutive quartets of all 0s** can be replaced with a double colon (::)

2001:0DB8:0000:0000:0000:0000:0080:34BD

↓

2001:0DB8::0080:34BD

↓ Combine both methods

2001:DB8::80:34BD

But, there's a limitation here.

18:36 / 39:25
IPv6 Address Abbreviation

Shortening (abbreviating) IPv6 addresses

Full IPv6 Address	Shortened IPv6 Address
2000:AB78:0020:01BF:ED89:0000:0000:0001	2000:AB78:20:1BF:ED89::1
FE80:0000:0000:0000:0002:0000:0000:FBE8	FE80::2:0:0:FBE8
AE89:2100:01AC:00F0:0000:0000:0000:020F	AE89:2100:1AC:F0::20F
2001:0DB8:8B00:1000:0002:0BC0:0D07:0099	2001:DB8:8B00:1000:2:BC0:D07:99
2001:0DB8:0000:0000:0000:0000:0000:1000	2001:DB8::1000

You're able to replace five quartets of 0s with the double colon.

Expanding shortened IPv6 addresses

Full IPv6 Address	Shortened IPv6 Address
FE80:0000:0000:0000:1010:02FC:0000:0009	FE80::1010:2FC:0:9
2001:0DB8:0001:0B23:2309:0000:0000:00C1	2001:DB8:1:B23:2309::C1
FD00:0000:0000:0000:1000:0689:9000:0CDF	FD00::1000:689:9000:CDF
FF02:0000:0000:0000:0000:0000:0000:0002	FF02::2
0000:0000:0000:0000:0000:0000:0000:0001	::1

I will talk about different IPv6 address types
in another video, but each of these addresses

Finding the IPv6 prefix (global unicast addresses)

- Typically, an enterprise requesting IPv6 addresses from their ISP will receive a /48 block.
- Typically, IPv6 subnets use a /64 prefix length.
- That means an enterprise has 16 bits to use to make subnets.
- The remaining 64 bits can be used for hosts.

2001:0DB8:8B00:0001:0000:0000:0000:0001/64

48-bit 'global routing prefix' assigned by the ISP

16-bit 'subnet identifier', used by the enterprise to make various subnets

However, that doesn't mean you'll only ever see /64 prefix lengths.

64-bit 'interface identifier', the host portion of the address

Finding the IPv6 prefix

52 56

16 32 48

300D:00F2:0B34:2100:0000:0000:1200:0001/56

↓

300D:F2:B34:2100::/56

~~300D:F2:B34:21::/56 = 300D:00F2:0B34:0021::/56~~

So, remember that point, you can only remove the 'leading' 0s.

Finding the IPv6 prefix

16 32 48 64 80 84 88 92

2001:0DB8:8B00:0001:FB89:017B:0020:0011/93

0d11

0b1011 → 0b1000

0d8

0x8

2001:DB8:8B00:1:FB89:178::/93

I hope you can see the importance of really understanding binary.

Finding the IPv6 prefix

Host Address	Prefix
FE80:0000:0000:0000:4c2c:e2ed:6a89:2a27/9	FE80::/9
2001:0DB8:0001:0B23:BA89:0020:0000:00C1/64	2001:DB8:1:B23::/64
2001:0DB8:0BAD:CAFE:1300:0689:9000:0CDF/71	2001:DB8:BAD:CAFE:1200::/71
2001:0DB8:0000:FEED:0DAD:018F:6001:0DA3/62	2001:DB8:0:FEEC::/62
2001:0DB8:9BAD:BABE:0DE8:AB78:2501:0010/63	2001:DB8:9BAD:BABE::/63

Note that you don't have to write out the shortened version, you can write out the whole

27:28 / 39:25
Identifying the IPv6 Prefix

Global unicast addresses

- **Global unicast** IPv6 addresses are public addresses which can be used over the Internet.
- Must register to use them. Because they are public addresses, it is expected that they are globally unique.
- Originally defined as the `2000::/3` block (`2000::` to `3FFF:FFFF:FFFF:FFFF:FFFF:FFFF:FFFF:FFFF`).
- Now defined as all addresses which aren't reserved for other purposes.

`2001:0DB8:8B00:0001:0000:0000:0000:0001/64`

48-bit 'global routing prefix' assigned by the ISP

16-bit 'subnet identifier', used by the enterprise to make various subnets

64-bit 'interface identifier', the host portion of the address

global unicast address. Global routing prefix, subnet identifier, and interface identifier.

Unique local addresses

- **Unique local** IPv6 addresses are *private* addresses which **cannot be used over the Internet.**
- You do not need to register to use them. They can be used freely within internal networks and don't need to be globally unique (*). Can't be routed over the Internet.
- Uses the address block `FC00::/7` (`FC00::` to `FDFF:FFFF:FFFF:FFFF:FFFF:FFFF:FFFF:FFFF`)
- However, a later update requires the 8th bit to be set to 1, so the first two digits must be `FD`.
- *The global ID should be unique so that addresses don't overlap when companies merge.

FD45:93AC:8A8F:0001:0000:0000:0000:0001/64

Indicates a unique local

16-bit 'subnet identifier', used by the enterprise to make various subnets

64-bit 'interface identifier', the host portion of the address

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1 month free

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- *The global ID should be unique so that addresses don't overlap when companies merge.

`FD45:93AC:8A8F:0001:0000:0000:0000:0001/64`

Indicates a unique local address

40-bit 'global ID', which should be randomly generated

16-bit [illegible] used by the enterprise to make various subnets

Then the last 64 bits are the interface ID, just like global unicast addresses.

64-bit 'interface identifier', the host portion of the address

13:40 / 39:32 Unique Local addresses

Link local addresses

- **Link-local** IPv6 addresses are automatically generated on IPv6-enabled interfaces.

- Use command `R1(config-if)#` **`ipv6 enable`** on an interface to enable IPv6 on an interface.

- Uses the address block `FE80::/10` (`FE80::` to `FEBF:FFFF:FFFF:FFFF:FFFF:FFFF:FFFF:FFFF`)
- However, the standard states that the 54 bits after `FE80/10` should be all 0, so you won't see link local addresses beginning with `FE9`, `FEA`, or `FEB`. Only `FE8`.

- The interface ID is generated using EUI-64 rules.

- *Link-local* means that these addresses are used for communication within a single link (subnet). Routers **will not** route packets with a link-local destination IPv6 address.

- Common uses of link-local addresses:
  → routing protocol peerings (OSPFv3 uses link-local addresses for neighbor adjacencies)
  → next-hop addresses for static routes
  → *Neighbor Discovery Protocol* (NDP, IPv6 replacement for ARP) uses link-local addresses to function

I'll talk about NDP as well as IPv6 routing in Day 33, but let me show you a small example.

Multicast addresses

- **Unicast** addresses are one-to-one.
  ↳ One source to one destination.

- **Broadcast** addresses are one-to-all.
  ↳ One source to all destinations (within the subnet).

- **Multicast** addresses are one-to-many.
  → One source to multiple destinations (that have joined the specific *multicast group*).

- IPv6 uses range `FF00::/8` for multicast. (`FF00::` to `FFFF:FFFF:FFFF:FFFF:FFFF:FFFF:FFFF:FFFF`)

- **IPv6 doesn't use broadcast** (there is no 'broadcast address' in IPv6!)

to send a message to all hosts in the subnet using multicast, which I'll show you in the next slide.

18:28 / 39:32
Multicast addresses